# MUHAMMAD ADNAN

AI/ML Intern | AI Engineer | Agentic AI, RAG & LLM Application Developer
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Portfolio | LinkedIn | GitHub

## PROFESSIONAL SUMMARY

Computer Science graduate and current AI/ML Intern at XICTEK Systems, specializing in agentic AI, LLM application development, and AI-driven automation. Hands-on experience designing and deploying LLM-powered applications and multi-agent systems using LangChain, LangGraph, and CrewAI, integrating OpenAI, Anthropic, and Google Gemini APIs for production use cases. Skilled in building RAG pipelines (vector embeddings, ChromaDB), computer vision pipelines (CNN/MobileNetV2, OpenCV) for real-time image classification, and connecting AI systems to APIs, webhooks, and databases to automate end-to-end business processes. Comfortable with prompt engineering, structured outputs, tool/function calling, and AI evaluation, with additional experience in voice/conversational AI and full-stack delivery (Python, FastAPI, Next.js, Docker).

## CORE SKILLS

**AI & Agents:** LLM Applications, Agentic Workflows, LangChain, LangGraph, CrewAI, OpenAI / Anthropic / Google Gemini APIs

**RAG & AI Pipelines:** RAG, Vector Embeddings, ChromaDB, Intelligent Data Extraction, Classification, Summarization, Enrichment, AI Evaluation

**Computer Vision & ML:** CNN, MobileNetV2, OpenCV, Scikit-learn, TensorFlow, Keras, Model Fine-Tuning & Evaluation

**Automation & Integration:** Workflow Automation, REST APIs, Webhooks, Tool/Function Calling, Structured Outputs, CRM Integrations

**Conversational AI:** Voice AI Agents, Speech-to-Text, Conversational AI, Voice-Based Customer Interaction

**Full-Stack & Deployment:** Python, FastAPI, Next.js, React, React Native, Django, Docker, Railway, Firebase

**Data:** Pandas, NumPy, SQL, MySQL, Power BI, DAX

## EXPERIENCE

**AI/ML Intern — XICTEK Systems (Remote)** Oct 2026 – Present

- Building AI-powered applications using Python, FastAPI, and Large Language Model (LLM) APIs such as Gemini, including a chatbot with a working demo and GitHub repository.
- Applying prompt engineering, API integration, and error handling to build reliable, user-friendly AI solutions.
- Working in a remote, structured team environment using GitHub, ClickUp, and Microsoft Teams, with regular implementation reviews and clear project documentation.

**Full-Stack Web Developer Intern — DevSynt (Rise Summer Internship 2026)** Summer 2026

- Completed the DevSynt Rise Summer Internship 2026 in Full Stack Web Development and received a Certificate of Completion (ID: DS-RISE-2026-C1-BA347D0C).
- Worked across the full-stack development workflow using Next.js, React, FastAPI, and MySQL, with Git/GitHub for version control.

**Business Associate (Founder) — Scroll'n Ecom, Rawalpindi, Pakistan** Jan 2025 – Apr 2026

- Founded and managed an e-commerce startup, overseeing product sourcing, online store management, digital marketing, and customer relations.
- Applied data-driven decisions to pricing, marketing spend, customer engagement, and day-to-day business operations.

## SELECTED AI & AUTOMATION PROJECTS

**Nexora AI — Agentic AI Commerce Intelligence Platform**

*Agentic AI, LLMs, RAG, Voice AI, Multi-Agent Orchestration (LangGraph), FastAPI*

- Built an AI-powered e-commerce intelligence platform using multi-agent orchestration (LangGraph) to automate business analysis and operational decision-making.
- Integrated a Voice AI Agent for conversational customer interaction and AI-powered support, extending the platform beyond text-based workflows.
- Designed RAG-powered AI copilot capabilities delivering intelligent insights for demand forecasting, inventory, pricing, promotions, logistics, and customer support.

**AI-Based Fertilizer Deficiency Monitoring Application**
*CNN / MobileNetV2, OpenCV, React Native, Django, Firebase*

- Developed an AI-powered mobile application for real-time NPK deficiency detection in rice and maize leaf images, achieving 96.3% classification accuracy.
- Built an end-to-end pipeline from image capture and CNN inference to REST API delivery and actionable fertilizer recommendations.

**AI-Powered AutoML SaaS Platform**
*FastAPI, Next.js 14, Python, Gemini API, ChromaDB, SHAP, Docker, Railway*

- Built a full-stack SaaS platform that automatically trains and benchmarks seven ML models per uploaded dataset and ranks them on a live leaderboard.
- Integrated a dataset-aware AI chatbot and RAG-powered documentation assistant using ChromaDB + Gemini API to answer questions from the project's codebase and documentation.
- Added automated data-quality analysis, explainability, report generation, a no-code API generator, RBAC, Docker containerization, and production deployment on Railway.

**Job Search Assistant — Multi-Agent CrewAI System**
*CrewAI, Gemini API, SerpAPI, Streamlit, Python*

- Built and deployed a 7-agent CrewAI pipeline powered by Gemini to automate job discovery, matching, and summarization, with a dark-themed Streamlit UI and search filters.
- Debugged SerpAPI integration and agent hallucination issues to improve the reliability and accuracy of agent outputs.

## EDUCATION

**B.Sc. Computer Science — Pir Mehr Ali Shah Arid Agriculture University** 2022 – 2026
Rawalpindi | CGPA: 3.39 / 4.0 | Completed

**F.Sc. Pre-Engineering — Zubair Public School & College, Dera Ismail Khan** Completed 2022
Grade: A

## CERTIFICATIONS

- Full Stack Web Development — DevSynt Rise Summer Internship 2026 (Certificate of Completion), 25 September 2026
- Data Science Orientation — IBM (via Coursera), 18 November 2025
- GenAI Job Simulation — BCG X (Forage), August 2026
- Automation AI Accelerator: From Co-pilot to Autonomous Agent Job Simulation — Datacom (Forage), August 2026